



Investigating the Effects of Partial Agonism at TAAR1 on Ethanol-Sucrose Consumption

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Abstract

Alcohol use disorder has a high prevalence, health and social cost, and few pharmacological treatments available. Trace amine associated receptor 1 (TAAR1) modulates dopaminergic activity in key reward pathways and can decrease consumption of substances of abuse in rodent models. The present study investigated the effects of TAAR1 activation on ethanol consumption in male and female Long Evans rats using the partial agonist RO5263397. It was hypothesized that RO5263397 would reduce ethanol consumption and preference following a 9-week self-administration training period and after two weeks of abstinence. A two-bottle paradigm was used: subjects could self-administer an ethanol-sucrose mixture, or water. Ethanol consumption (g/kg body weight) and preference (from total fluid) were measured at baseline, under a dose of 10mg/kg (i.p.) RO5263397, and under vehicle (0.9% v/v saline, i.p.) in all subjects. RO5263397 decreased ethanol consumption in both experiments (before and after abstinence), but overall preference for ethanol-sucrose did not change across conditions due to a parallel decrease in water consumption. Female subjects drank more and had a greater decrease in consumption than males when treated with RO5263397. A novelty suppressed feeding task assessed anxiety and locomotor activity: subjects tested with RO5263397 were quicker to reach a food reward than those treated with the vehicle, discouraging the explanation that RO5263397 decreased locomotion. Several explanations are explored to mitigate these unexpected findings, and future avenues suggested. TAAR1 remains a promising target in alcohol use disorder, with proven efficacy in reducing consumption and few detrimental side effects.

Characterizing the Effects of Partial Agonism at TAAR1 on Ethanol-Sucrose Consumption

Harmful alcohol use has profound impacts at individual and societal levels, with alcohol-related harm costing the taxpayer \$9.1 billion annually in New Zealand (NZIER, 2024). Alcohol is a widely accessible drug, and many individuals drink without issue. Yet, alcohol has strong abuse potential, with many individuals suffering from Alcohol Use Disorder (AUD), a condition characterized by the compulsive seeking and use of alcohol despite negative health or social consequences (Carvalho et al., 2019). Nutt and colleagues (2010) named alcohol as the most harmful drug, taking the prime position above heroin and methamphetamine. As it stands, AUD has few effective pharmacological treatments: disulfuram and naltrexone have shown mixed to modest clinical efficacy (Ray et al., 2018). It is imperative that modern pharmacological treatments are explored alongside behavioural interventions to mitigate the incidence and impact of AUD.

Theories of Addiction

All drugs of abuse, including ethanol, acutely enhance dopamine in the mesocorticolimbic circuitry, neural pathways associated with reward processing. This circuitry comprises dopaminergic projections from the ventral tegmental area (VTA) to the nucleus accumbens (NAc), prefrontal cortex, amygdala, and hippocampus. Robinson and Berridge's (1993) Incentive Sensitization Theory suggests that repeated exposure to a substance of abuse causes hyper-sensitization of this circuitry to drug-associated cues, increasing the feeling of 'wanting' the substance, without enhancing the 'liking' or pleasure of use. Further, Koob and Le Moal's (2004) Allostasis Theory suggests that exposure initially increases dopamine, but subsequent neuroadaptation over time lowers baseline dopamine: higher doses are required to achieve the initial euphoria. Regulation of dopamine in the mesocorticolimbic circuitry offers a promising therapeutic strategy.

Trace Amine Associated Receptor 1

In 2001, two independent research groups (Bunzow et al., 2001; Borowsky et al., 2001) reported the discovery of a family of trace amine receptors that modulated major monoaminergic systems. The most studied, trace amine associated receptor 1 (TAAR1), has been widely implicated in the psychiatric disease literature since its discovery. Drugs targeting the receptor have proven effective at ameliorating a range of symptoms of psychiatric disease and have recently reached large-scale clinical trials in humans. Further, the receptor has been studied in some addiction literature, as TAAR1 agonists have consistently reduced consumption of amphetamines (Pei et al., 2017). However, the role of TAAR1 in alcohol use disorder remains understudied, presenting an opportunity for translatable pharmacological treatment if TAAR1 activation can reduce ethanol consumption.

Most psychopharmacological treatments act on monoamine systems directly and can cause serious side effects due to the widespread roles of monoamine receptors (Gonda et al., 2023). For example, all approved schizophrenia pharmacotherapies target the dopamine 2 receptor (D2R), causing side effects like motor inhibition and Parkinson's-like symptoms (Kuvarzin et al., 2014). Ulotaront is a full agonist at TAAR1 that was developed as a novel antipsychotic and does not act directly upon D2 receptors. The compound recently underwent late-stage clinical trials for schizophrenia, and showed minimal side effects (Dedic et al., 2021). Yet, despite reducing both positive and negative symptoms, the novel compound did not significantly improve symptoms above a large placebo effect, which masked its benefits (Otsuka, 2023). Drug development is a lengthy and expensive process, so additional uses of candidate compounds like ulotaront could provide accessible options for treating AUD (Litten et al., 2016).

TAAR1 is a G-protein coupled receptor: activation of G_s and G_q proteins causes downstream action via cyclic AMP/protein kinase A (cAMP/PKA) and calcium/protein

kinase C (Ca⁺⁺/PKC pathways), resulting in phosphorylation of PKA and PKC (Lynch et al., 2013). These cascades lead to the phosphorylation of the dopamine transporter (DAT), a protein that regulates the reuptake of dopamine and controls extracellular dopamine levels, thereby dampening dopamine-associated reinforcement. Furthermore, TAAR1 interacts with dopamine D2 receptors (D2R) through heteromerization (Espinoza et al., 2011). Antagonism of the D2R enhances TAAR1 signaling, and TAAR1 signaling is required for D2 receptor-mediated auto-inhibition of dopamine release (Houdant et al., 2025). Both TAAR1 and D2 are required to maintain homeostatic regulation of dopamine activity. Regulating dopamine transmission in the reward circuitry decreases the reinforcing effects of rewarding stimuli, presenting a promising avenue for addiction treatment.

TAAR1 is enriched in dopaminergic regions associated with reward, including the ventral tegmental area (VTA), nucleus accumbens, substantia nigra, and dorsal and ventral striatum (Rutigliano, 2018). TAAR1 is activated by monoamines, such as dopamine and serotonin, as well as endogenous trace amines. Trace amines, like octopamine, tryptamine, and phenylethylamine, are biochemically similar to neurotransmitters but are present in low concentrations in the brain (Bunzow et al., 2001). In TAAR1 knockout animals, where the gene is rendered non-functional, the behavioural repertoire is almost identical to that of wildtype subjects, but knockout animals are hypersensitive to dopamine, highlighting the role of TAAR1 in dopamine modulation (Espinoza et al., 2015).

TAAR1 and Addiction

In a seminal study of TAAR1 and alcohol consumption, Lynch et al. (2013) used transgenic mouse models to investigate the effects of a *Taar1* gene knockout (TAAR1-KO) on alcohol preference when compared to wildtype (TAAR1-WT) in male and female mice. In a two-bottle choice paradigm (2BC), where animals could access and self-administer ethanol and water, TAAR1-KO mice had a higher preference for ethanol (3% v/v to 11% v/v ethanol)

than TAAR1-WT mice. Lynch and colleagues conducted a parallel experiment assessing sucrose preference: TAAR1-KO mice showed no significant difference from TAAR1-WT mice in their preference for sucrose over water, suggesting that the effect was ethanol-specific. The authors concluded that the increased consumption of ethanol in knockout mice demonstrated TAAR1's role in modulating the reinforcing effects of ethanol. Although genetic models are fundamentally distinct from pharmacological interventions, these findings implicated TAAR1 in the consumption of addictive substances, providing the foundations for neuropharmacological research to build upon.

Frycz and colleagues (2023) recently investigated the effects of TAAR1 activation on ethanol consumption and preference in female mice using a full agonist, RO5256390. In a 2BC protocol, mice were statistically classified into 'high' and 'low' drinkers and treated with RO5256390 (10mg/kg, intraperitoneally). The authors utilized a progressive ratio schedule where mice had to perform an increasing number of nose pokes to access alcohol, a measure of motivation to access alcohol. High drinking mice had significantly lower consumption of ethanol and performed fewer nose pokes when treated with RO5256390 than when treated with a vehicle. 'Low drinkers' showed no significant difference in ethanol consumption or motivation in response to RO5256390. Frycz and colleagues also examined the effect of TAAR1 activation following a 9-day period of alcohol abstinence. 'High drinkers' treated with RO5256390 consumed significantly less alcohol following abstinence than those treated with the vehicle, yet there was no significant difference in alcohol consumption for 'low drinkers'. These results imply that the benefits of TAAR1 activation depend on one's propensity for alcohol, leaving us to hypothesize that the high/low-drinker distinction may be indicative of an underlying neuroadaptation that is effectively ameliorated by TAAR1 activation. However, this conclusion requires a better operationalization of addiction, where subjects must overcome a negative stimulus to access ethanol. The authors'

post-abstinence results are of particular interest due to the relapse liability seen in AUD and warrant further investigation of TAAR1 activation following abstinence.

Houdant and colleagues (2025) compared partial and full agonism at TAAR1 on drinking behaviour and found distinct mechanisms. Using partial (RO5263397: vehicle, 1mg/kg) and full (RO5166017: vehicle, 15mg/kg) agonists, and male C57BL/6JRj mice, the authors measured ethanol (20% v/v) consumption and preference in a 2BC paradigm, and NAc dopamine levels. Both compounds decreased ethanol consumption and preference, with the partial agonist showing a greater decrease in consumption than the full agonist (43% vs 25% reduction). Surprisingly, the two compounds exhibited distinct mechanisms: full agonism increased dopamine release in ethanol-consuming mice, whereas partial agonism had the opposite effect, resulting in a decrease in dopamine release in ethanol-consuming mice. In alcohol-naïve mice, the full agonist reduced dopamine release while the partial agonist showed no significant change. These results suggest a distinction in the action of partial and full agonists, differing also in the level of substance exposure, which again supports the hypothesis that TAAR1 activation mitigates the underlying neuroadaptations from drug exposure, and encourages further investigation of partial agonism at TAAR1.

Wu and colleagues (2020) investigated the effects of the partial agonist RO5263397 on behavioural sensitization, as measured by ethanol-induced hyperlocomotion, in male and female mice. The behavioural sensitization procedure involved administration of ethanol (2.4 mg/kg) daily for 10 days, thought to reflect the ongoing neuroadaptations caused by exposure to a substance of abuse. The authors found that exposure to RO5263397 prevented the development of behavioural sensitization to ethanol and reduced the expression of behavioural sensitization in both males and females. This finding was dose-dependent and relied upon the presence of TAAR1, as it was absent in knockout animals. Behavioural sensitization has less face validity than a 2BC protocol: animals are injected with ethanol

over a period of 10 days, rather than accessing it of their own volition. The 2BC is more representative of how individuals access substances of abuse.

Many TAAR1 studies have used rodent models including C57BL/6J mice, an inbred strain commonly used in addiction science due to their preference for ethanol, and various rat strains, which willingly self-administer ethanol. Behavioural diversity is required to ensure representativeness of the interindividual differences seen in humans, where some individuals are more prone to developing addiction than others (Swendson & Le Moal, 2011). There are also well-established sex differences in ethanol consumption: females are quicker to establish and escalate drug-taking than males and are more likely to relapse after a period of abstinence (Becker et al., 2016). Yet, sex differences in behavioural response to TAAR1 activation are not currently elucidated: no known studies have investigated the impact of TAAR1 activation on ethanol consumption in females compared with males.

Our Study

The present study aims to elucidate the effects of a partial agonist at TAAR1, RO5263397, on alcohol consumption and preference in male and female Long Evans rats, an outbred strain with substantial interindividual differences in ethanol propensity. Both male and female rats will be included to assess sex differences in treatment outcomes. As in Frycz and colleagues' study where high-drinkers were more impacted by activation at TAAR1, we expect that females will show a greater reduction in ethanol consumption under RO5263397 due to their high consumption levels. Animals will be trained to self-administer ethanol using a two-bottle choice paradigm (2BC) over a 'habituation period' with an intermittent access schedule. Inconsistent access to ethanol induces high levels of consumption in Long Evans rats (Simms et al., 2008). Baseline consumption will be measured and all subjects be tested under both drug conditions (vehicle, 10mg/kg RO5263397) with counterbalancing. We expect that RO5263397 will significantly decrease alcohol consumption and preference for

alcohol in comparison with saline. A novelty suppressed feeding task, where animals can reach a food reward in a novel anxiogenic environment, will be conducted to assess any locomotor deficits and anxiolytic effects of RO5263397. A two-week abstinence from ethanol will allow assessment of the effects of TAAR1 agonism on relapse-like behaviour: we expect that RO5263397 will reduce post-abstinence consumption and preference.

Method

Subjects

Twenty-four Long Evans adolescent rats were obtained from Te Herenga Waka – Victoria University small animal facility at 6 weeks old (post-natal day 42). This comprised of 12 males (initial weight: $M = 330\text{g}$, $SD = 23\text{g}$) and 12 females (initial weight: $M = 227\text{g}$, $SD = 11\text{g}$). Subjects were housed in same-sex pairs for the duration of the study, excluding during 4-hour ‘drinking sessions’, in which they were housed individually. Cages were ventilated and lit under a 12-hour light/dark cycle where the lights turned off at 9am: temperature ($20^{\circ}\text{C} \pm 1^{\circ}\text{C}$) and pressure (60% humidity) were controlled. All drinking sessions and experiments were conducted under red light and dark conditions as is consistent with the popular Drinking in the Dark (DiD) paradigm which makes use of rodents’ circadian rhythms to increase voluntary ethanol consumption (Morales et al., 2020). All subjects had *ad libitum* access to food and water, excluding during drinking sessions as the drinking cages did not have food access to rule out any confounding effects of food on drinking behaviour. Animals were monitored daily for signs of distress or illness. All animal procedures were conducted in accordance with guidelines and approved by Victoria University Animal Ethics Committee (2025/AE040006, see Appendix A).

Materials

Ethanol-sucrose solutions were prepared by adding sucrose (Fischer Chemical, UK) 5% w/v and ethanol 99.8% (Fischer Chemical, UK) diluted to 10% (v/v) ethanol in room-

temperature tap water for a two-bottle choice paradigm. 100 mL (milliliters) of ethanol-sucrose was placed in the ethanol-sucrose bottles, and 100mL tap water in designated water bottles. All bottles were leak-proof, and were weighed before and after drinking sessions to determine consumption in mL. Animals were weighed weekly to ascertain body weight. Body weight was used to determine the consumption of ethanol in grams per kilogram body weight (g/kg). TAAR1 partial agonist RO5263397 was obtained in powder form from Hoffmann-La Roche Ltd (Switzerland). RO5263397 dosage (10mg/kg) was selected based on robust findings (Frycz et al., 2023 [0, 10 mg/kg]; Espinoza et al., 2018 [0.1, 1.0, 10mg/kg]). The vehicle used was saline (0.9% v/v). Doses were delivered intraperitoneally (i.p.) at a volume of 1mL/kg. Order of administration was based on a within-subject Latin Square design and drug administrations were separated by at least 72-hours.

Study Timeline

Table 1

Timeline of Experimental Protocol Across 15-Week Study Duration

Week	Procedure	Purpose
1, 2, 3, 4, 5	Daily adaptation to ethanol and sucrose at varying concentrations (10%, 5%, 0%)	Allow animals to acclimatize to environment and handling
6, 7, 8	Daily access, two-bottle choice: 10% ethanol + 5% sucrose versus tap water	Increase consumption to establish highest consumption patterns
9	IA2BC (MWF), baseline consumption measurements across 4 drinking sessions	Establish baseline fluid intake (water & ethanol mix)
10	RO5263397 & saline injections on Monday (counterbalanced), consumption tests in 2BC	Experiment 1 protocol
11	RO5263397 & saline injections on Wednesday, consumption tests in 2BC	Experiment 1 protocol
12	IA2BC with 10% ethanol, 5% sucrose	Re-establish baseline fluid consumption
13, 14	Abstinence from ethanol; novelty suppressed feeding task (NSFT) with	Two-week abstinence period; Measuring locomotor activity with NSFT

	RO5263397 & saline (between-subjects)	
15	RO5263397 & saline injections, counterbalanced with 72-hours between, consumption tests in 2BC	Experiment 2 protocol. End of experiments

Note. Procedures were performed during the light phase (0900–2100 h). All animals were single housed during testing periods. Week 0 = PND 42; IAB2C: Intermittent Access 2 Bottle Paradigm; MFW = Monday, Wednesday, Friday; NSFT = Novelty Suppressed Feeding Task

Design

Intermittent Access Two Bottle Choice Paradigm

On ‘drinking days’, water bottles were removed from subjects’ home cages as the lights turned off (9am). For the drinking sessions, the subjects were placed in individual cages for four hours, during which they had access to two bottles: one with tap water and one with the ethanol-sucrose solution, a standard two-bottle choice (2BC) procedure. Sucrose was added to make the solution more palatable, and a sucrose fading procedure allowed a decrease from 10 % w/v sucrose to 5% w/v sucrose. The fading procedure aimed for 0% sucrose; however, most subjects ($n = 17$) did not continue drinking to biologically relevant levels (drank $< 1\text{mL}$ of 10% v/v ethanol) when sucrose was removed from the solution, necessitating the use of 5% (w/v) sucrose throughout the experiments. The solution used throughout the experiment was 10% ethanol (v/v) and 5% sucrose (w/v), from here described as ethanol-sucrose. During the first 6 weeks, subjects had intermittent access to the 2BC (IA2BC) paradigm, accessing alcohol on Mondays, Wednesdays, and Fridays, as seen in Table 1. The habituation took longer than expected, with daily access to ethanol-sucrose (during Weeks 6 through 8) to drive ethanol-sucrose consumption. The side that ethanol-sucrose was presented on was switched at each drinking day to control for side bias. Once drinking had stabilized in (did not significantly increase or decrease) for one week, baseline consumption levels (ethanol-sucrose and water) were measured for three days and averaged (Week 9), and the testing period began. Table 1 below shows the experimental summary.

Animals were assigned into 2 groups ($n = 12$ in each group) for counterbalancing: on each experimental day, one group had vehicle (saline) and the other group had RO5263397. During experimental days (separated by 9 days), the subjects were injected (intraperitoneally) with RO5263397 or the vehicle and placed into the 2BC paradigm within 15 minutes (in accordance with Dorotenko et al., 2020). Following Experiment 1, the subjects had schedule one week (Week 12) of IA2BC to re-establish baseline drinking. Following this re-establishment, a 2-week abstinence period occurred, in which the subjects had no access to ethanol-sucrose. In Experiment 2, consumption and preference were measured using the same dose of the TAAR1 agonist and vehicle, counterbalanced for group membership (different randomization than Experiment 1) with a 72-hour delay between sessions.

High vs Low-Drinkers

Following the protocol set out by Frycz et al. (2023), animals were assigned to be ‘high-drinkers’ or ‘low-drinkers’ depending on their baseline level of consumption. Following Frycz’s protocol, 30% of subjects ($n = 4$ per subgroup) who had the highest baseline ethanol consumption (g/kg) were classified as ‘high drinkers’ while the remaining animals were classified as ‘low drinkers’. This dimension was treated as a categorical variable (high/ low) that was used as a covariate in post-hoc analysis.

Novelty Suppressed Feeding Task

During the two-week abstinence period (Week 13), all subjects were introduced to a small cereal (Fruit Loop, ~2 g) and had access daily for four days. On the fifth day (during the first week of ethanol abstinence), a novelty suppressed feeding task (NSFT) was conducted over a 5-minute period to measure the exploratory behaviour of each subject. The test was completed in a black Perspex arena (0.8m x 0.8m) under red light conditions with a single Fruit Loop in the middle of the arena. Animals were deprived of food for ~ 2 to 5 hours prior to the NSFT. In each trial, the rat was placed facing the wall in one corner and

exploratory behaviour was recorded using Ethovision XT. Latency to eat was recorded. Half of the subjects were injected with 10mg/kg i.p. RO5263397 ($n = 12$) and the other half ($n = 11$) were injected with the vehicle (0.9% v/v saline, i.p.): group membership was randomized based on a Latin Square design. Subjects began the task 20 minutes following dosage, as in line with pharmacological evidence (Espinoza et al., 2018). Due to tracking software issues, locomotion data was unavailable.

Data Analysis

All statistical analyses were conducted using R (Version 4.5.1, 13 June 2025) and RStudio (Posit, 2025, v5.1.513.3). Statistical significance was set at $\alpha = .05$ for all analyses. Ethanol consumption was determined as a function of body weight (g/kg) at each time point. Water consumption (in mL) was measured during each drinking session and a preference ratio was determined (each subject's preference for the ethanol-sucrose solution over total fluid).

For Experiment 1, mixed ANOVAs were conducted to compare the three levels of condition (within-subjects: baseline, saline, RO5263397) and sex (between-subjects: male, female) on ethanol consumption (g/kg), water consumption (mL) and preference for ethanol-sucrose over total fluid (water + ethanol/sucrose). For Experiment 2, mixed ANOVAs were conducted with a 2 x 2 design (sex x condition) with the same dependent variables (ethanol consumption in g/kg, consumption of water in mL, and proportion of preference for the ethanol-sucrose mixture over total fluid). A 2 (condition) x 2 (high/low drinker classification) x 2 (sex) design was used to analyse the effects of high/low classification on ethanol consumption (g/kg).

Mixed-design ANOVAs with condition as a within-subjects variable and sex and/or high/low classification as a between-subjects variable are most appropriate for repeated

measures experimental designs (Altman & Krzywinski, 2015). For the NSFT, where both condition and sex were between-subjects variables, a between-subjects factorial ANOVA was conducted to test main effects, following standard ANOVA protocol for comparing means (Krzywinski & Altman, 2014). *Post-hoc* power analyses were conducted when results gained significance. Bar graphs were created in R using the ‘ggplot’ package to visualize group differences and replicated in Excel (v16.79.1). Error bars in all plots represent $\pm$ SEM. Effect sizes were reported as partial n^2 , or generalized eta-squared (*ges*), depending on the analysis, and interpreted based on conventions by Lakens (2013).

Exclusions

Data from one subject (Subject 17, female) was removed from water consumption and preference analyses due to a leaky water bottle leading to inaccurate data in both Experiments 1 and 2 (although data from this subject for g/kg ethanol consumption and NSFT were unimpacted). Unfortunately, one subject (Subject 1, male [M]) was lost following Experiment 1 due to premature death. This occurred two days after Experiment 1 and is not expected to have impacted the subject’s consumption. All subsequent analyses (NSFT, post-abstinence RO5263397 treatment) excluded this participant. During post-abstinence experiments, there were several more instances of leaky bottles resulting in loss of data for subjects 6 (M), 13 (female [F]), and 23 (F) in Experiment 2, leaving a final sample size of $N = 20$ for Experiment 2.

Results

Experiment 1

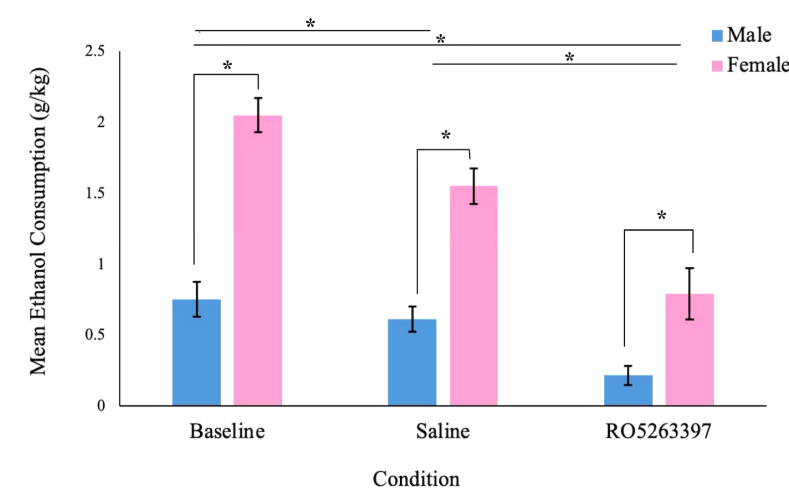
Ethanol Consumption

A mixed-ANOVA for ethanol consumption across sex and condition demonstrated a significant main effect of sex, $F_{(1, 22)} = 41.95$, $p < .001$, $ges = .568$, and a significant main

effect of condition, $F_{(1.91, 42.21)} = 58.7, p < .001, ges = .453$. There was also a significant interaction effect, $F_{(1.91, 42.12)} = 9.23, p < .001, ges = .115$. A Greenhouse-Geisser correction was made due to a violation of the assumption of sphericity assumption. On average, females showed higher ethanol consumption in g/kg across conditions (Baseline: $M = 2.05, SD = 0.42$; Saline: $M = 1.55, SD = 0.44$; Drug: $M = 0.79, SD = 0.63$) compared to males (Baseline: $M = 0.75, SD = 0.42$; Saline: $M = 0.61, SD = 0.31$; Drug: $M = 0.22, SD = 0.24$). Pairwise comparisons (Tukey-corrected) indicated that all means differed significantly from each other (all $p < .05$), except for the difference between baseline and saline in male subjects ($p = .40$). These findings are demonstrated in Figure 1.

Figure 1

Mean Ethanol Consumption (g/kg) by Condition and Sex



To rule out the possibility that Condition x Sex interaction effect was derived from the non-significant difference between baseline and saline conditions in males, a 2 (saline, drug) $\times$ 2 (male, female) mixed-design ANOVA was conducted to examine differences in ethanol consumption (g/kg) in only the saline and drug conditions. There was a significant main effect of Sex, $F_{(1, 22)} = 24.09, p < .001, ges = .457$, where females consumed significantly more than males overall, and a significant main effect of Condition, $F_{(1, 22)} = 46.77, p < .001, ges = .329$, with subjects consuming less ethanol under RO5263397 than under saline.

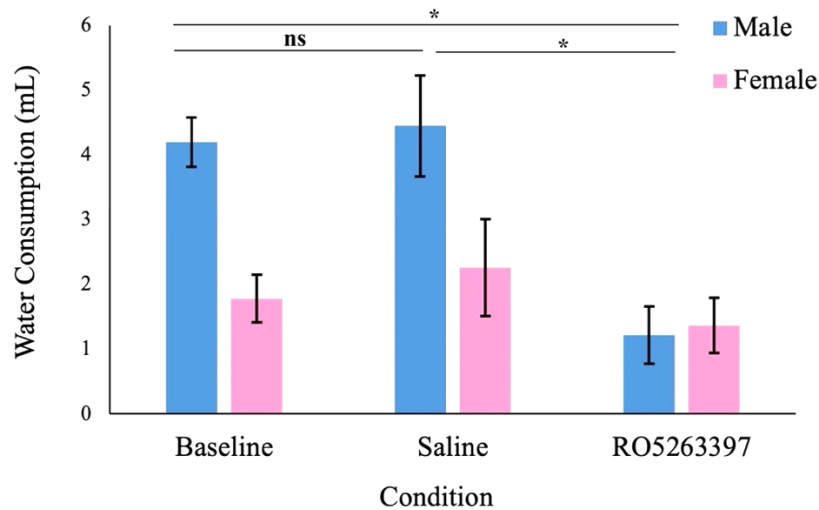
Importantly, there was still a significant Condition $\times$ Sex interaction, $F_{(1, 22)} = 4.61, p = .043$, $ges = 0.05$, although with a smaller effect size. Simple means analyses showed that the drug condition reduced consumption more strongly in females than in males.

Water Intake Across Conditions

A mixed ANOVA was conducted to compare water consumption (mL) by Sex and Condition. As before, a Greenhouse-Geisser correction was made due to a violation of the assumption of sphericity assumption, and all pairwise comparisons were Tukey-adjusted. Analysis of water intake indicated a significant main effect of Sex, $F_{(1, 21)} = 14.14, p = .001$, with males ($M = 3.29, SD = 2.57$) consuming more water overall than females ($M = 1.81, SD = 1.76$). There was also a significant main effect of Condition, $F_{(1.6, 33.6)} = 7.07, p = .005$, with no difference in consumption from Baseline ($M = 2.99, SD = 1.21$) to Saline ($M = 3.36, SD = 2.44$) and a significant decrease in the RO5263397 condition ($M = 1.29, SD = 1.43$). Pairwise comparisons indicated no significant difference in consumption between Baseline and Saline ($p = .804$), but the decrease from Baseline to RO5263397 was statistically significant ($p = .0029$) and the decrease from Saline to RO5263397 was also significant ($p = .021$). The interaction between Sex and Condition did not reach significance, $F_{(1.6, 33.6)} = 2.96, p = .076$, although visual inspection suggested a stronger decrease in water consumption under RO5263397 in males than females (see Figure 2).

Figure 2

Water Consumption (mL) by Condition and Sex



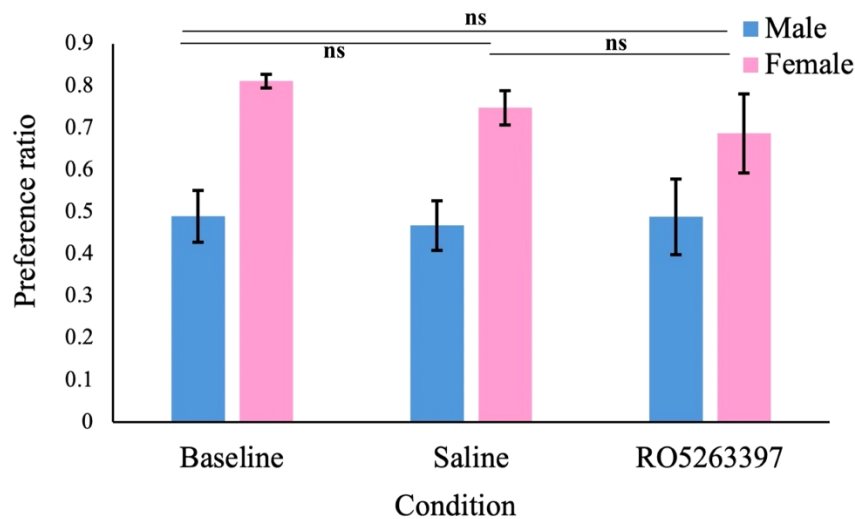
Ethanol Preference Across Conditions

A mixed ANOVA was conducted to compare ethanol-sucrose preference across Condition and Sex. As above, the assumption of sphericity was violated ($p < 0.05$), so Greenhouse-Geisser corrected F values were used. The main effect of Sex was significant, $F_{(1, 21)} = 15.09, p < .01$, with females demonstrating a significantly higher preference for ethanol ($M = 0.75, SD = 0.20$) than males ($M = 0.48, SD = 0.24$). There was no main effect of Condition, $F_{(1.47, 30.90)} = 0.65, p = .48$, nor was there a significant interaction between Condition and Sex, $F_{(1.47, 30.90)} = 0.63, p = .49$. These results support the null hypothesis that preference did not change across conditions, and the pattern of change across conditions was consistent for both males and females. Due to the lack of statistical significance, no pairwise comparisons were conducted. These results are plotted in Figure 3.

Analysis of counterbalancing used a 2 (condition) x 2 (group: drug first vs saline first) mixed ANOVA. Results indicated no significant main effect of Group, $F_{(1, 21)} = 0.06, p = .806$, no significant main effect of Condition, $F_{(1, 21)} = 0.13, p = .718$, and no significant Group x Condition interaction term, $F_{(1, 21)} = 0.23, p = .634$.

Figure 3

Preference for Ethanol-Sucrose over Total Fluid in Each Condition and Sex



Total Fluid Consumption

A 2 (Condition) x 2 (Sex) ANOVA was conducted to examine sex differences in total fluid consumption (ethanol-sucrose + water, in mL). Following a Greenhouse-Geisser correction, there was a significant main effect of Condition, $F_{(1, 21)} = 26.89, p < .001, ges = 0.453$, such that fluid consumption was higher at Baseline ($M = 8.71, SD = 2.12$) and Saline ($M = 8.05, SD = 3.62$) than under RO5263397 administration ($M = 3.39, SD = 2.09$). However, there was no significant main effect of Sex, $F_{(1.59, 33.36)} = 1.50, p = 0.234, ges = .025$, and no Sex x Condition interaction effect, $F_{(1.59, 33.36)} = 0.88, p = .402, ges = .026$.

High/Low Drinkers

A 2 (Condition) x 2 (Sex) x 2 (Drinking Classification: High, Low) ANOVA was conducted to examine differences in ethanol consumption (g/kg) across conditions in high and low drinkers. As expected, there was a significant main effect of Sex, a main effect of Condition, a main effect of High/Low Classification, and a significant interaction effect of Sex and High/Low Classification. There was no interaction between Classification and Condition, $F_{(1, 20)} = 0.02, p = .89$, and there was no significant three-way interaction, $F_{(1, 20)} = 0.822, p = .38$.

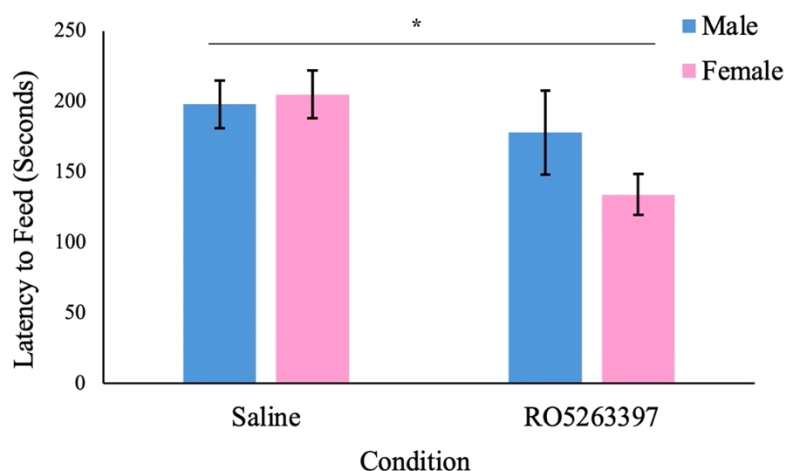
Novelty Suppressed Feeding Task

A 2 (condition: saline, RO5263397) x 2 (sex: male, female) between-subjects ANOVA was conducted to determine the effects of sex and condition on latency to eat the food reward (in seconds). The main effect of condition was significant, $F_{(1, 19)} = 5.06$, $p = .04$, $\eta p^2 = .24$, suggesting that 24% of variance in latency was explained by RO5263397 treatment, a large effect size (Lakens, 2013). There was no significant main effect of sex, $F_{(1, 19)} = 0.723$, $p = .406$, $\eta p^2 = .11$, nor was there a significant interaction effect between condition and sex, $F_{(1, 19)} = 1.53$, $p = .232$, $\eta p^2 = .07$. Sex did not affect latency, and the drug effect did not differ between males and females. A *post-hoc* power analysis of the main effect of condition indicated that the achieved power was 56%. A predictive power analysis suggested that a total sample of $N = 40$ (20 per group) would be required to reach 80% power for an effect of this size.

A simple linear regression was conducted to examine whether testing order predicted latency to feed in the novelty-suppressed feeding task. The model did not gain significance, $F_{(1, 21)} = 0.05$, $p = .831$, $R^2 = .002$, suggesting the order of testing did not impact latency.

Figure 4

Mean Latency on Novelty Suppressed Feeding Task



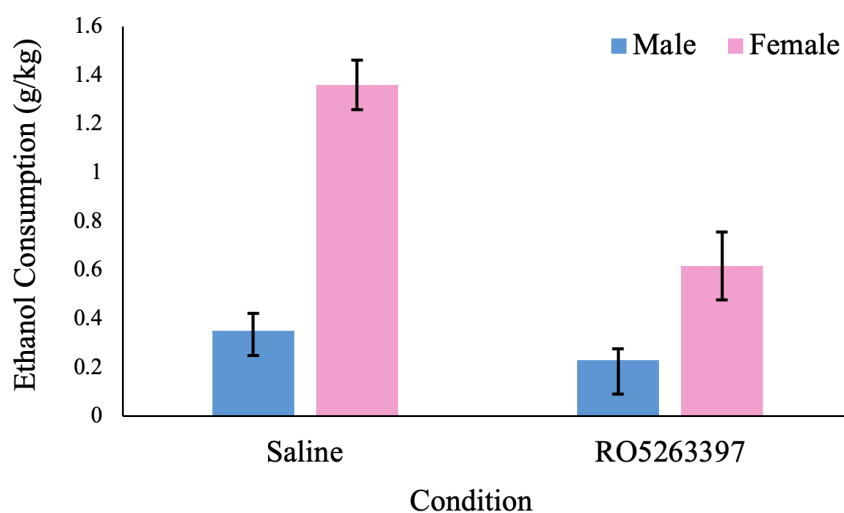
Experiment 2

Ethanol Consumption

A 2 (condition: saline, RO5263397) x 2 (sex) mixed ANOVA was conducted to compare ethanol consumption in g/kg following abstinence. There was a significant main effect of condition, $F_{(1, 19)} = 38.54, p < .001$: animals consumed significantly less ethanol under RO5263397 administration compared to saline. There was a significant main effect of sex, $F_{(1, 19)} = 39.79, p < .001$: females consumed significantly more than males. Mean ethanol consumption ($\pm$ SEM) for males was 0.351 ± 0.07 g/kg in the saline condition and 0.229 ± 0.05 g/kg in the RO5263397 condition. For females, mean consumption was 1.36 ± 0.10 g/kg (saline) and 0.617 ± 0.14 g/kg (RO5263397). There was also a significant interaction effect between sex and condition, $F_{(1, 19)} = 20.53, p < .001, \eta^2 = 0.227$, suggesting that females had a greater reduction in consumption under RO5263397 than did males. *Post-hoc* pairwise comparisons indicated all pairwise comparisons were significant excluding the decrease from saline to RO5263397 (0.122 g/kg ethanol) in males ($p = .213$).

Figure 5

Mean Ethanol Consumption in g/kg Body Weight by Condition and Sex

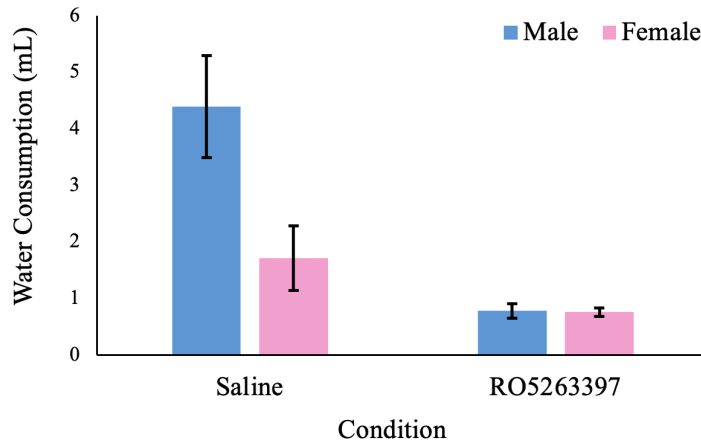


Water Consumption

Another mixed ANOVA was conducted to examine differences in water consumption by Condition and Sex. There was a significant main effect of Sex, $F_{(1, 17)} = 6.31, p = .024, \eta^2 = .15$, indicating that overall water consumption was higher for males ($M = 2.59, SD = 1.63$) than for females ($M = 1.23, SD = 0.97$). There was a significant main effect of Condition, $F_{(1, 17)} = 16.12, p = .008, \eta^2 = .41$, indicating that overall water consumption was higher in the Saline condition ($M = 3.05, SD = 2.29$) than in the drug condition ($M = 0.77, SD = 0.317$). There was also a significant Condition and Sex interaction effect on water consumption, $F_{(1, 17)} = 5.45, p = .03, \eta^2 = 0.14$; males had a greater reduction in water consumption under RO5263397 than did females. *Post-hoc* pairwise comparisons were conducted using estimated marginal means which used model-based adjustments to account for exclusions which had left an uneven number of males ($n = 10$) and females ($n = 9$) and Bonferroni's correction for multiple comparisons. The difference between Saline and RO5263397 (3.61mL) for males was significant ($p < .001$), but the difference of 0.96mL for females did not reach statistical significance ($p > .05$). The difference in consumption between males and females in the Saline condition (2.68mL, $p = .002$) was statistically significant, but not in the RO5263397 condition (0.02mL, $p = .98$).

Figure 6

Mean Water Consumption (mL) Across Condition and Sex

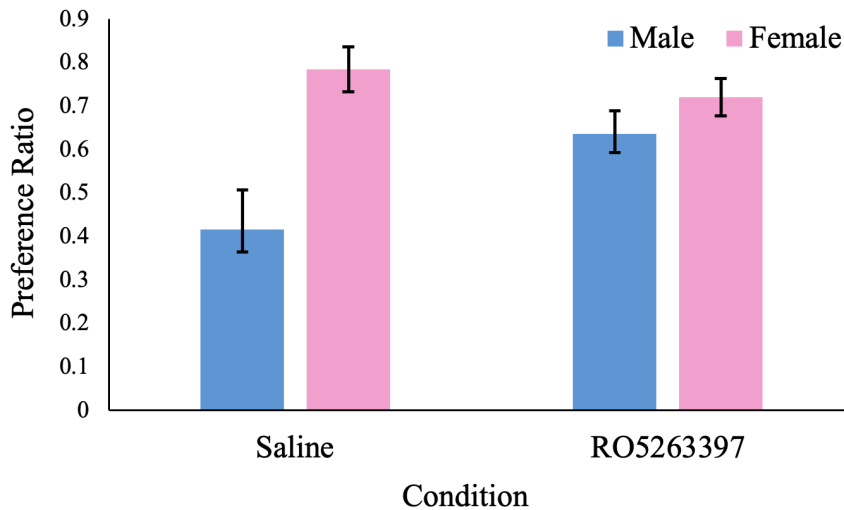


Ethanol Preference

Again, using a mixed ANOVA design for ethanol preference across Condition and Sex, we found no significant main effect of Condition, $F_{(1, 17)} = 2.33, p = .145, ges = .04$. A significant main effect of sex was found, $F_{(1, 17)} = 9.02, p = .008, ges = .27$, with females showing higher proportion scores than males ($M = 0.73, SD = 0.14; M = 0.53, SD = 0.23$, respectively). The Condition $\times$ Sex interaction gained significance, $F_{(1, 17)} = 7.86, p = .012, ges = .13$, indicating that the effect of condition differed between males and females with a medium effect size. Follow-up pairwise comparisons revealed that males showed a significant increase in preference of 0.22 (22%) from saline to RO5263397 ($p < .001$), whereas females did not (6%, $p = .39$). Females had a significantly higher preference than males in the saline condition ($p = .003$), but there was no difference in preference across sex in the drug condition ($p = 0.25$). A *post-hoc* power analysis in R yielded a low power of 19%; a predictive power analysis suggested that 33 subjects were needed to gain power of 80%.

Figure 7

Ethanol Preference Following Abstinence by Condition and Sex



Analysis of counterbalancing for preference in Experiment 2 used a 2 (Condition) x 2 (Group: drug first vs saline first) mixed ANOVA. Due to missingness, group sizes were unequal (drug first: $n = 11$; saline first: $n = 9$). Results indicated no significant main effect of Group, $F_{(1, 18)} = 0.63, p = .437$, no significant main effect of Condition, $F_{(1, 18)} = 3.81, p = .067$, and no significant Group x Condition interaction term, $F_{(1, 18)} = 2.01, p = .173$.

Total Fluid Consumption

A 2 (Condition) x 2 (Sex) ANOVA was conducted to examine sex differences in total fluid consumption (ethanol-sucrose + water, in mL). There was a significant main effect of Condition, $F_{(1, 17)} = 42.33, p < .001, ges = 0.545$, such that fluid consumption was higher under Saline ($M = 7.11\text{mL}, SD = 2.34$) than under RO5263397 administration ($M = 2.87\text{mL}, SD = 1.62$). There was no significant main effect of Sex, $F_{(1, 17)} = .79, p = 0.384, ges = .024$, and no Sex x Condition interaction effect, $F_{(1, 17)} = 0.22, p = .646, ges = .006$.

High/Low Drinkers

Another mixed ANOVA (Condition x Sex x High/Low Classification) was conducted to examine the effects of RO5263397 on ethanol consumption in g/kg post-abstinence, using

the same high/low classification (classification based upon consumption levels prior to either experiments starting). As expected, there were significant main effects of Sex, Condition, high/low classification, and an interaction between Sex and Condition. However, there was no significant interaction of Sex and High/Low Classification, $F_{(1, 17)} = 3.00, p = .102$, nor a significant interaction between Condition and Classification, $F_{(1, 17)} = 0.27, p = .611$. The three way interaction (between Sex, Condition, and High/Low Classification) did not reach statistical significance, $F_{(1, 17)} = 0.63, p = .440$.

Discussion

Summary of Findings

The present study investigated the effect of RO5263397 on ethanol consumption and preference in male and female Long Evans rats: as expected, RO5263397 decreased ethanol consumption (corrected for body weight) in both following initial habituation, and after two weeks' abstinence. This supports our primary hypothesis and aligns with surrounding literature (Houdant et al., 2025). However, our additional findings highlight a complex relationship between RO5263397 and consumption: due to a parallel decrease in water consumption under RO5263397, there was no significant decrease in preference for ethanol-sucrose over water in any condition. In Experiment 2, males had reduced preference in the saline condition, an unexpected finding. In the NSFT, RO5263397-treated animals were quicker to eat than their saline-treated peers, suggesting no detrimental impact of RO5263397 on general locomotion. Our findings point to the broad influence of the reward system on behaviour and illustrate the difficulty in distinguishing between addictive behaviours and general reward sensitivity.

Sex Differences

As expected, females drank more and had a higher preference for ethanol-sucrose than did males in almost all analyses (non-significant difference in preference under RO5263397 in Experiment 2). This finding is consistent with a wealth of earlier literature demonstrating this pattern (Nieto & Kosten, 2017). For ethanol consumption (g/kg), a significant interaction effect existed between sex and condition: females exhibited a greater reduction in consumption than males under RO5263397. This was not the case for ethanol-sucrose preference in Experiment 1: there was no interaction between Condition and Sex on preference. However, in Experiment 2, RO5263397 significantly *increased* ethanol-sucrose preference in males, with a medium effect size, but did not significantly change preference in females. Our study found no significant sex differences in total liquid consumption per condition in either Experiment 1 or 2, suggesting these effects were specific to ethanol-sucrose consumption.

The present study is the first to report a sex difference in ethanol self-administration in response to RO5263397. Wu and colleagues (2020) found an overall effect of condition on ethanol behavioural sensitization in response to RO5263397 (0.1 and 0.32mg/kg), yet did not compare sexes. Similarly, Lynch and colleagues (2013) investigated ethanol consumption in both males and female TAAR1 knockout and wildtype mice, yet made no sex comparisons. Our findings demonstrate a significant difference between sexes in response to RO5263397 on ethanol-sucrose consumption, but not total fluid consumption. Furthermore, there was no significant main effect of sex on latency to feed in the NSFT, suggesting the sex difference in response to RO5263397 may be limited to ethanol-sucrose consumption. Further investigation of these sex differences is warranted.

Novelty Suppressed Feeding Task

The NSFT is used to assess anxiety: a longer latency indicates heightened anxiety in a novel environment (Samuels & Hen, 2011). Animals treated with RO5263397 had a lower

mean latency to reach a food reward in the NSFT than subjects treated with saline. There was no main effect of sex, and no interaction effect between sex and condition. Decreased latency in the RO5263397 group is consistent with evidence of the anxiolytic effects of TAAR1 activation, with ulotaront currently in clinical trials for the treatment of generalized anxiety disorder (Li et al., 2025). However, the NSFT is not a ‘pure’ measure of anxiety: latency also depends on appetite and locomotion. Increased appetite under RO5263397 does not explain our findings: no evidence suggests that RO5263397 increases hunger, and there is some evidence that TAAR1 activation reduces feeding behavior (Dedic et al., 2024). Differences in locomotion may have impacted the NSFT finding, and our study could not isolate anxiety from locomotion due to the lack of tracking data. However, the surrounding literature suggests that RO5263397 does not reduce locomotion (10mg/kg; Thorn et al., 2014). The most plausible explanation for the decreased latency finding was the anxiolytic effect of RO5263397.

High/Low Classification

Post-hoc analysis using the High/Low Drinker classification as employed by Frycz and colleagues (2023) was not informative: there was no significant interaction between condition and classification in either experiment, discouraging the hypothesis that high drinkers would show a greater reduction in consumption in response to RO5263397 than low drinkers. This is likely due to the arbitrary classification of high/low drinkers: our subjects did not fall into two clusters. Rather than further statistical manipulations, a better experimental design would enhance the representativeness of interindividual differences in addictive profiles. This may involve an assay of neuroadaptation following repeated ethanol exposure using fast-scan cyclic voltammetry (FSCV; Liu et al., 2018).

Saline Administration

Surprisingly, the administration of saline reduced ethanol consumption from baseline in Experiment 1 but only in females; for males, saline did not significantly decrease ethanol or

water consumption from baseline. This finding may be explained by the stress of the intraperitoneal injection: female subjects may have been more distressed from the injection, thus suppressing their drinking. However, there were no sex differences in the NSFT. These results may be better understood using a biological assay of anxiety as a covariate, allowing differences in consumption to be linked to states of physiological arousal. Further investigation of the impact of sex differences in response to acute stressors on drinking behavior could include an acute stressor like an electric foot shock, commonly used in rodent models (Bali & Jaggi, 2015).

Potential Mechanisms and Explanations

Horizontal and Vertical Locomotion

One might dissociate the vertical movement required to reach the water bottles in the 2BC paradigm (~12.5cm, requiring subjects to stand) from the horizontal movement required in the NSFT. A particularly relevant study by Sukhanov and colleagues (2019) investigated the response to RO5263397 in compulsive behaviour using schedule-induced polydipsia (SIP), a symptom of schizophrenia, in Wistar rats. For animals with SIP, RO5263397 decreased fluid consumption and vertical activity (fewer 15cm-upwards movements) without reducing horizontal movement. Further, these findings were limited to high doses (6 and 10mg/kg) and were not present at 1-3mg/kg. In subjects without SIP, high doses decreased water consumption (vertical activity was not measured). These results align directly with ours: the 10mg/kg dose may have reduced vertical movement (2BC task) but not horizontal movement (NSFT task), while reducing general consumption. We did not measure vertical locomotion so no further conclusions can be drawn but adaptations of the present study should include vertical oscillations. Furthermore, testing lower doses may elucidate the threshold for reduction in vertical movement, allowing us to characterize the relationship between movement and fluid consumption under RO5263397.

Thirst Versus Hunger

We may also make a distinction between the consumption of fluids (water, ethanol-sucrose) and solids (Fruit Loop), leading us to hypothesize that RO5263397 decreased thirst but not hunger. Emerging evidence suggests that TAAR1 regulates metabolic processes. A recent study by Dedic and colleagues (2024) found that RO5263397 (0.3 mg/kg) modulated neuronal activity in key homeostatic brain regions including hypothalamic centers, the amygdala, and appetite-associated brainstem nuclei, concluding that TAAR1 plays a central role in energy homeostasis. We may hypothesize that TAAR1 activation impacts homeostatic regulation in the hypothalamus, leading to a general reduction in consumption. Although TAAR1 is expressed in the hypothalamus (Rutigliano et al., 2018), no known studies have investigated its role in this region. It is well established that D2 receptors contribute to dopaminergic stimulation of the hypothalamic-pituitary-adrenal axis (Borowsky & Kuhn, 1992), and D2R form heteromers with TAAR1 (Espinoza et al., 2011). The role of TAAR1 activation and the interaction between D2 and TAAR1 in the hypothalamus is a future avenue for study to elucidate the mechanism through which general fluid consumption decreases.

General Motivation

The sucrose preference test, a 2BC paradigm of sucrose and water, is a typical test of anhedonia in rodents: deficits in this task are thought to reflect a reduction in the ability to experience reward (Primo et al., 2023). The observed reduction in ethanol-sucrose consumption in the present study may suggest a wider affective or motivational deficit under RO5263397. However, the NSFT finding does not support a general motivational deficit explanation. Furthermore, recent evidence from Dorotenko and colleagues (2020) suggested that RO5263397 (0.3-6mg/kg) has pro-cognitive benefits: rats improved in a sustained attention task at higher doses, and there was some evidence implicating TAAR1 in inhibitory behavioural control. Adaptations may rule out anhedonia as an alternative explanation: a

progressive ratio schedule, where animals must produce an increasing amount of nose pokes to access ethanol, would determine each animal's 'breakpoint' (where they stop responding) as a measure of reward valuation.

Conditioned Taste Aversion

Liu and colleagues (2022) found that RO5263397 (3.2 and 5.6mg/kg) induced conditioned taste aversion to saccharin and table salt in rats without reducing total fluid consumption. This is particularly relevant to the present study: the reduction in consumption may arise from RO5263397 making the ethanol-sucrose solution aversive. However, the results of the NSFT do not support a taste aversion explanation: animals treated with RO5263397 showed a reduced latency to reach the palatable food. Further, the surrounding literature suggests that TAAR1 activation is specific to ethanol and does not reduce sucrose consumption (Lynch et al., 2023). Clinical trials of ulotaront provided no reports of taste aversion and side effects were minimal (Le et al., 2023). Furthermore, emerging evidence of the effect of TAAR1 activation on metabolic processes supports the hypothesis that TAAR1 activation can control metabolic processes and reduce food intake (Raab et al., 2016).

Limitations and Future Directions

The use of sucrose throughout the ethanol self-administration protocol hinders our capacity to parse out the highly addictive effects of ethanol from the naturally reinforcing sucrose and restricts the conclusion that the reduction in consumption is specific to ethanol. Further, our findings surrounding sex differences are limited as females prefer both ethanol and sucrose to a greater degree than males (Grimm et al., 2022). However, the literature suggests that TAAR1 activation shows specificity and does not reduce sucrose consumption. For example, Lynch and colleagues' (2013) knockout study found that TAAR1-KO animals showed greater ethanol consumption and preference than wildtype animals yet demonstrated no difference in sucrose preference in a 2BC paradigm. Further, Cotter and colleagues (2015)

found that full TAAR1 agonist RO5203648 (0, 3, 10mg/kg) dose-dependently reduced methamphetamine self-administration in a response-lever paradigm without impacting sucrose self-administration in rats. Despite the use of sucrose, our findings highlight the capacity of RO5263397 to reduce consumption of rewarding substances.

Another obvious limitation is the lack of an ‘addiction’ model: the animals did not overcome a negative stimulus to access the ethanol-sucrose mixture. Further, our 2-week abstinence period may not have caused withdrawal. A better operationalization of ‘addiction’ might include a mixture with quinine, an aversive substance, alongside ethanol to categorize animals who continue drinking despite the unpleasant taste. Translation to alcohol use disorder in humans is limited by our simple methodology: animals were socially isolated for four hours with no entertaining stimuli. This is not representative of the social complexity of human drinking, where the context one finds themselves in substantially impacts consumption (Sudhinaraset et al., 2016). Our method could be improved with more sophisticated equipment: Frycz et al. (2023) used IntelliCages where mice could remain in social cages with behavioural monitoring, allowing conclusions that are more representative of naturalistic drinking patterns.

As Houdant and colleagues (2025) demonstrated, RO5263397’s effects on dopamine release relied upon prior ethanol exposure: RO5263397 increased dopamine release in ethanol-naïve animals yet decreased dopamine release in rats that had been repeatedly exposed to ethanol. Dopamine reductions in regions like the nucleus accumbens, within the mesocorticolimbic circuitry, have a greater impact than in other areas. Further studies may benefit from an assay of dopamine expression levels in key reward regions at baseline compared with RO5263397 treatment using FSCV.

There are endless hypotheses that could be explored to reconcile the present findings. TAAR1 plays a modulatory role throughout the central nervous system, so our findings could be considered in a wide-reaching manner (Rutigliano et al., 2018). The use of multiple doses

(0.1, 1, 3 mg/kg) and hence creation of a dose response curve might better elucidate whether the decrease in fluid consumption is found across lower doses. We may also assess consumption and locomotion hourly throughout the 4-hour drinking session due to the 3-hour half-life of RO5263397 (Thorn et al., 2014).

Conclusions

Our findings, although limited in some respects, support the hypothesis that TAAR1 activation can ameliorate alcohol use disorder. RO5263397, a partial agonist at TAAR1, successfully reduced consumption of a highly rewarding substance, ethanol-sucrose. We have proposed several avenues for investigation of the role of TAAR1 in vertical locomotor activity, metabolic homeostasis, and anxiety, all potentially impacting the present study. As Liu and colleagues (2024) recently highlighted, the behavioural and chemical effects of TAAR1 have been widely studied, yet few studies have investigated the cellular mechanisms through which these results arise. Future work should consider the complex behavioural impact of partial agonism, as well as the cellular mechanisms that underlie the benefits of TAAR1 activation. This will allow characterization of TAAR1 action throughout the brain and reduce off-target effects. As the field continues to move towards large clinical trials, a cautionary approach must be employed: the primary goal in treating alcohol use disorder should be to enhance functioning universally, rather than to replace addiction with affective or motivational flattening.

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Appendix A



Animal Ethics Approval Certificate

Project Number: 2025/AE040006

Project Title: Targeting the trace amine-associated receptor 1 (TAAR1) for the treatment of alcohol use disorder: A rat model

Version: 1.01

Chief Investigator (CI): Prof Juan Canales
Victoria University of Wellington

Other Participants:

Associate Investigator
Prof Albertus Ellenbroek

Honours Student Rosalind Walker

Approval Committee: Victoria University of Wellington Animal Ethics Committee

Approval Duration: 01 May 2025 to 01 May 2026

Approval date: Friday, 9 May, 2025

Dear Prof Juan Canales

Your application to use live animals in your research was approved by the VUW Animal Ethics Committee on [Meeting date]. Approval was given for the requested period of [Approval duration], expiring on [Expiry date].

We thank you for your cooperation in helping the Committee ensure appropriate animal welfare standards and procedures are in place at Victoria University of Wellington.



Peter Pfeffer AEC Chair

<https://www.wgtn.ac.nz/research/support/ethics/animal-ethics>

School of Biology

Proviso (*If applicable*)

Appendix B

Standard Operating Procedures Details

SOP 69P Rat Injections

This SOP summary describes the procedures to be followed for injections in rats – please view the long version for see full details

Needle and syringe size

A 24 gauge (24 G) or smaller needle is used to minimise tissue trauma and discomfort. In general, 27 G needles are used, but a different size can be appropriate depending on the circumstances:

- A smaller needle (29 or 30 G) to inject pups;
- A bigger needle (24 or 25 G) is used if the solution to inject is viscous and cannot be easily injected.
- For the injection of electrolytes post-surgery (10 mL), when the rat is still anaesthetised, 22 G needles can be used. This size must not be used if the animal is conscious.

1 mL Luer Lock® syringes are commonly used as most injection volumes are under 0.5 mL and the Luer Lock tip reduces the risk of leakage and splatter in case of high backpressure. Very low volumes (less than 0.1 mL, to inject pups for example) can be delivered from ultra-fine insulin syringes (0.3 mL), while higher volumes (> 1 mL) should be delivered from a suitable syringe (3 mL, 5 mL or 10-20 mL) to avoid multiple injections.

Preparation

General:

- Minimise the handling time to minimise stress
- The solution must be as close as possible to pH 7, not cold, and as small as practical.
- Use sterile technique;
- Keep a record of the injection date and volume.

Preparation:

- Weigh each rat and write down the weight and volume to inject (based on the body weight).
- Prepare the syringe and needle with the required volume, expel any air bubbles
- After injection, watch for signs of distress and pain. Laboured breathing and hunched posture might indicate that internal organs got hit by the needle.
- When several injections are done in one session, or an injection is given several days in a row (manipulation grade C), alternate sites should be used with each injection

Subcutaneous (s.c.) injection

- Restrain the rat appropriately
- Raise the skin to create a fold either in the scruff of the neck, the flank or the lower back, then insert the needle in the base of the fold. Make sure the needle has gone through the skin (as opposed to into the dermis) but hasn't punctured out through both layers of skin, or doesn't hit the muscle underneath the skin as this is also painful to the animal.
- Administer the solution in one smooth motion, gently, but not too slowly. If a blockage is felt, or extra force is needed on the plunger, it is likely because the tip of the needle is in a bad location (in the dermis or the muscle); adjust the needle position until the solution goes freely.
- Watch for any fluid coming out directly from the point of injection: if a drop of solution appears on the skin, it is likely to be leaking from the injection site (while injecting, or after injection if a big needle is used); do not administer more solution but make a note of it. Depending on the volume injected, some puffiness can appear at the injection site; it usually clears in less than 30 min.

Intraperitoneal (i.p.) injection

Only consider the i.p. administration route if the s.c. route cannot be used, because of the high rate of i.p. miss-injection (10-20 % of i.p. injections end up in the gut, the bladder or behind the peritoneal space) and more challenging restrain techniques.

Maximum injection volume: 5-10 mL in an adult rat.

- Restrain the rat appropriately:
 - o Insert the needle, preferably in the lower-right quadrant of the rat's abdomen at a 60 degree angle against the body to avoid damaging the cecum and other organs (that are usually located on the left). Be careful not to inject too close to the sulcus between the leg and the torso (can cause severe pain). If the animal urinates immediately after or during the injection, it is possible that the injection was made in the bladder.

Administer the solution in one smooth motion, gently, but not too slowly. If a blockage is felt, or extra force is needed on the plunger, it is likely because the tip of the needle is in a bad location; adjust the needle position until the solution goes free.

